ABSTRACT

Accurate and real-time prediction of transient stability is essential for ensuring the secure operation of power systems under large disturbances. The Transient Energy Function (TEF) method offers a fast and analytical way to assess stability by comparing system energy to a critical threshold, but its effectiveness depends heavily on precise knowledge of generator rotor angles and speeds. This project proposes an LSTM-based approach to enhance the accuracy of generator state prediction during fault conditions. By training a Long Short-Term Memory (LSTM) neural network on time-series simulation data from the IEEE 9-bus system, we aim to predict high-fidelity generator angles and speeds in real time. These predicted states are then integrated into the TEF framework to calculate kinetic and potential energies, thereby enabling more accurate estimation of the transient stability margin (ΔV). The proposed method is expected to outperform conventional approaches by providing more reliable stability assessment and improving control action decisions such as generation shedding.